

David Millard

Mechanical Engineering, Rochester Institute of Technology

US Citizen

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Education

- May. 2025-Present **Ph.D. in Mechanical Engineering**, Rochester Institute of Technology, NY, USA
- Sep. 2020-May. 2025 **B.S. in Computer Science**, Rochester Institute of Technology, NY, USA
GPA: 3.88/4
B.S. in Applied Statistics, Rochester Institute of Technology, NY, USA
GPA: 4/4
Project: Diffusion-augmented ensemble forecasting for initial condition perturbation of global atmospheric state.
- Jun. 2020 **Diploma in Mathematics and Science**, Mechanicville Jr/Sr Highschool, NY, USA. **GPA:** 3.92/4

Academic Projects

- ❖ Risk-aware **optimal transport** for distilled global models, Rochester Institute of Technology & University West
- ❖ Risk-aware **federated distributional agents** for complex private environments, Rochester Institute of Technology & University West
- ❖ Uncertainty quantification within function space using **conformal prediction** and **neural operators**, Rochester Institute of Technology & ETH Zurich
- ❖ Preconditioner learning via a dual optimization **deep reinforcement learning** algorithm, Rochester Institute of Technology
- ❖ **Diffusion**-augmented ensemble forecasting for initial condition perturbation, Rochester Institute of Technology & Lehigh University & Environment and Climate Change Canada
- ❖ V-PARADIS: a variational version of PARADIS for modelling **epistemic randomness**, Environment and Climate Change Canada
- ❖ PARADIS: **advection-diffusion** model for long term global weather state forecasting, Environment and Climate Change Canada
- ❖ An efficient **SWIN-Transformer** for atmospheric uncertainty quantification, Rochester Institute of Technology
- ❖ Adaptation of **LLMs** for the spread of misinformation, Rochester Institute of Technology
- ❖ **Vision language model** for analytic and numeric **PDE** solutions, Rochester Institute of Technology
- ❖ **Deep ritz** methods for elliptic differential-difference equations, Rochester Institute of Technology
- ❖ Data driven initial guess selection for **numerical weather prediction solvers**, Lehigh University & Environment and Climate Change Canada
- ❖ **Koopman operator** estimation for idealized atmospheric dynamics, Lehigh University & Environment and Climate Change Canada
- ❖ Abdominal trauma injury **segmentation** and **classification**, Rochester Institute of Technology
- ❖ Bird call classification on **EDGE compute constraints**, Rochester Institute of Technology

Publications

- ❖ **David Millard**, Lars Lindemann, Ali Baheri, Split Conformal Prediction in the Function Space with Neural Operators, Under review, 2025
 - ❖ **David Millard**, Arielle Carr, Stephane Gaudreault, Ali Baheri, Diffusion-augmented Ensemble Forecasting, Under review, 2025
 - ❖ **David Millard**, Arielle Carr, Stephane Gaudreault, Ali Baheri, PEARL: Preconditioner Enhancement through Actor-critic Reinforcement Learning, Under review, 2024
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- ❖ **David Millard**, Arielle Carr, Stephane Gaudreault, Deep Learning for Koopman Operator Estimation in Idealized Atmospheric Dynamics, 2024 IEEE Conference on Big Data
- ❖ **David Millard**, Arielle Carr, Stephane Gaudreault, Data-Driven Initial Guess Selection for Numerical Weather Prediction Solvers, 2024 IEEE Conference on Big Data
- ❖ Anton Selitskiy, **David Millard**, Deep Ritz Method for Elliptic Differential-Difference Equations, 2025 AMS Spring Easter Sectional Meeting

Awards and Honors

- ❖ IRES AI-PROWIL Cohort, Rochester Institute of Technology, USA, 2025
- ❖ SURF Research Fellowship, Rochester Institute of Technology, USA, 2025
- ❖ Undergraduate research scholar, Rochester Institute of Technology, USA, 2025
- ❖ Summa cum laude, Rochester Institute of Technology, USA, 2025
- ❖ Presidential Scholarship, Rochester Institute of Technology, USA, 2020-2025

Teaching Experience

- ❖ CSCI-720 Big data analytics, Rochester Institute of Technology, Spring 2024
- ❖ CSCI-335 Machine learning, Rochester Institute of Technology, Fall 2024-Spring 2024
- ❖ CSCI-420 Data mining, Rochester Institute of Technology, Spring 2024
- ❖ CSCI-721 Computer vision, Rochester Institute of Technology, Fall 2024
- ❖ Research assistant, Rochester Institute of Technology, USA, Spring 2024-Present

Research Experience

- ❖ Research assistant, Rochester Institute of Technology, Spring 2024
- ❖ Research experience undergraduate, Lehigh university, USA, Summer 2024
- ❖ Research assistant, Lehigh university, USA, Fall 2024-Spring 2025
- ❖ Research assistant, Environmental and climate change Canada, USA, Fall 2024-Spring 2025
- ❖ Research assistant, Rochester Institute of Technology, USA, Spring 2024-Present

Technical Skills

Software	❖ Python, OpenCV, PyTorch, TensorFlow, MATLAB and SIMULINK, C/C++, R, Java, C#, C++, Fortran, Cobal, Assembly
Stack	❖ Github, Agile, Waterfall, Trello
Hardware	❖ ARM microcontrollers
